

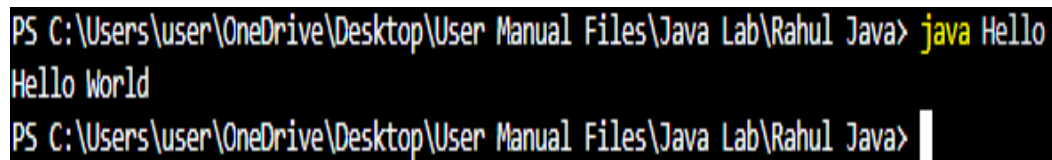
1. Write a program to display “Hello World” and execute it using the Command Line Interface (CLI) by manually setting the Path and ClassPath.

File Name: Hello.java

Source Code:

```
public class Hello {  
    public static void main(String[] args) {  
        System.out.println("Hello World");  
    }  
}
```

Output:



The screenshot shows a Windows Command Prompt window with a black background and white text. The prompt is at the C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java directory. The user has entered the command 'java Hello' and the output 'Hello World' is displayed on the next line. The prompt is ready for the next command.

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java Hello  
Hello World  
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```

2. Create a program that finds the second largest element in an array of 10 integers.

File Name: SecondLargest.java

Source Code:

```
import java.util.Scanner;

public class SecondLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[10];

        System.out.println("Enter 10 integers:");
        for (int i = 0; i < 10; i++) {
            arr[i] = sc.nextInt();
        }

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        for (int num : arr) {
            if (num > largest) {
                secondLargest = largest;
                largest = num;
            } else if (num > secondLargest && num != largest) {
                secondLargest = num;
            }
        }

        if (secondLargest == Integer.MIN_VALUE) {
            System.out.println("There is no second largest element.");
        } else {
            System.out.println("Second largest number: " + secondLargest);
        }
        sc.close();
    }
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac SecondLargest.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java SecondLargest
Enter 10 integers:
10 25 40 18 35 9 50 42 21 30
Second largest number: 42
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```

3. Demonstrate the use of a for-each loop to iterate through an array of Strings and print them in uppercase.

File Name: ForEachDemo.java

Source Code:

```
import java.util.Scanner;

public class ForEachDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String[] names = new String[5];

        System.out.println("Enter 5 names:");
        for (int i = 0; i < 5; i++) {
            System.out.print("Name " + (i + 1) + ": ");
            names[i] = sc.nextLine().trim();
        }

        System.out.println("\nNames in uppercase:");
        for (String name : names) {
            System.out.println(name.toUpperCase());
        }
        sc.close();
    }
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac ForEachDemo.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java ForEachDemo
Enter 5 names:
Name 1: Asur
Name 2: Bakasur
Name 3: Ravan
Name 4: Kans
Name 5: Bhoot

Names in uppercase:
ASUR
BAKASUR
RAVAN
KANS
BHOOT
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```

4. Write a program defining a Class and Object for a “Car” with attributes such as model and price, and a method to display them.

File Name: Car.java

Source Code:

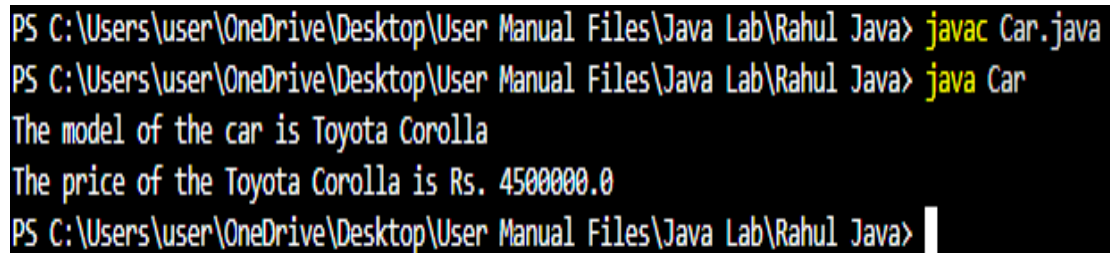
```
public class Car {
    String model;
    double price;

    Car(String model, double price) {
        this.model = model;
        this.price = price;
    }

    void display() {
        System.out.println("The model of the car is " + model);
        System.out.println("The price of the " + model + " is Rs. " + price);
    }

    public static void main(String[] args) {
        Car myCar = new Car("Toyota Corolla", 4500000.0);
        myCar.display();
    }
}
```

Output:



```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac Car.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java Car
The model of the car is Toyota Corolla
The price of the Toyota Corolla is Rs. 4500000.0
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```

5. Implement Constructor Overloading and Method Overloading in a class named Calculator to add integers and doubles.

File Name: CalculatorTest.java

Source Code:

```
class Calculator {
    Calculator() {
        System.out.println("Default Constructor");
    }

    Calculator(int value) {
        System.out.println("Value = " + value);
    }

    int add(int firstNumber, int secondNumber) {
        return firstNumber + secondNumber;
    }

    double add(double firstNumber, double secondNumber) {
        return firstNumber + secondNumber;
    }
}

public class CalculatorTest {
    public static void main(String[] args) {
        Calculator calculator1 = new Calculator();
        Calculator calculator2 = new Calculator(20);

        System.out.println("Integer Sum = " + calculator1.add(15, 25));
        System.out.println("Double Sum = " + calculator1.add(12.5, 7.5));
    }
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac CalculatorTest.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java CalculatorTest
Default Constructor
Value = 20
Integer Sum = 40
Double Sum = 20.0
```

6. Create a program to demonstrate the four access privileges (private, default, protected, and public) across two different packages.

File Name: Folder2/MyClass.java and Folder1/DemoClass.java

Source Code:

```
// File: Folder2/MyClass.java
package Folder2;

public class MyClass {
    private int a = 10;
    int b = 20;
    protected int c = 30;
    public int d = 40;

    public void show() {
        System.out.println("Private variable a = " + a);
        System.out.println("Default variable b = " + b);
        System.out.println("Protected variable c = " + c);
        System.out.println("Public variable d = " + d);
    }
}

// File: Folder1/DemoClass.java
package Folder1;

import Folder2.MyClass;

public class DemoClass {
    public static void main(String[] args) {
        MyClass obj = new MyClass();
        obj.show();
        System.out.println("Public variable accessed directly: " + obj.d);
    }
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac Folder2\MyClass.java Folder1\DemoClass.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java Folder1.DemoClass
Private variable a = 10
Default variable b = 20
Protected variable c = 30
Public variable d = 40
Public variable accessed directly: 40
```

7. Write a program that implements an Interface named RemoteControl with methods powerOn() and powerOff().

File Name: RemoteDemo.java

Source Code:

```
interface RemoteControl {
    void powerOn();
    void powerOff();
}

class TV implements RemoteControl {
    public void powerOn() {
        System.out.println("TV ON");
    }

    public void powerOff() {
        System.out.println("TV OFF");
    }
}

public class RemoteDemo {
    public static void main(String[] args) {
        TV t = new TV();
        t.powerOn();
        t.powerOff();
    }
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac RemoteDemo.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java RemoteDemo
TV ON
TV OFF
```

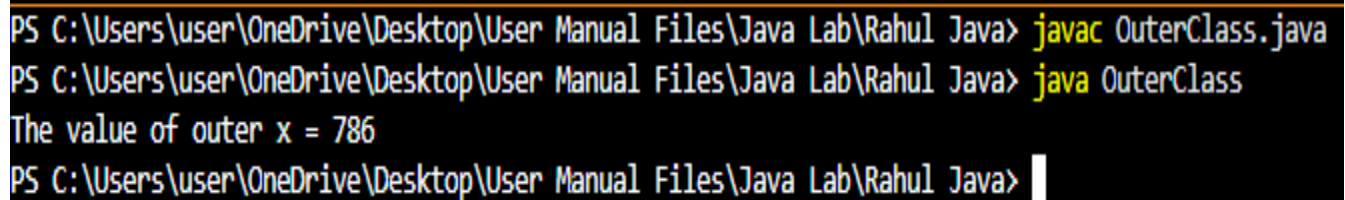
8. Define an Inner Class inside a class named Outer and demonstrate how to access the outer class members.

File Name: OuterClass.java

Source Code:

```
public class OuterClass {  
    int x = 786;  
  
    class InnerClass {  
        void show() {  
            System.out.println("The value of outer x = " + x);  
        }  
    }  
  
    public static void main(String[] args) {  
        OuterClass outer = new OuterClass();  
        OuterClass.InnerClass inner = outer.new InnerClass();  
        inner.show();  
    }  
}
```

Output:



```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac OuterClass.java  
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java OuterClass  
The value of outer x = 786  
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```


9. Create a program using the static keyword for a shared counter and the final keyword to create a constant value.

File Name: StaticFinalDemo.java

Source Code:

```
public class StaticFinalDemo {
    static int counter = 0;
    static final int MAX_VALUE = 100;

    StaticFinalDemo() {
        counter++;
    }

    public static void main(String[] args) {
        new StaticFinalDemo();
        new StaticFinalDemo();
        new StaticFinalDemo();

        System.out.println("Total objects created: " + counter);
        System.out.println("Constant value: " + MAX_VALUE);
    }
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac StaticFinalDemo.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java StaticFinalDemo
Total objects created: 3
Constant value: 100
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```

10. Demonstrate Multilevel Inheritance and show how the super keyword is used to call a parent constructor.

File Name: MultiLevelInheritance.java

Source Code:

```
class Vehicle {
    Vehicle(String brand) {
        System.out.println("Vehicle Brand: " + brand);
    }
}

class Car extends Vehicle {
    Car(String brand) {
        super(brand);
        System.out.println("Sports Car");
    }
}

class SportsCar extends Car {
    SportsCar(String brand) {
        super(brand);
        System.out.println("Sports Car Model: " + brand);
    }
}

public class MultiLevelInheritance {
    public static void main(String[] args) {
        new SportsCar("Ferrari");
    }
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac MultiLevelInheritance.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java MultiLevelInheritance
Vehicle Brand: Lamborghini
Car Type: Sports Car
Sports Car Model: Lamborghini
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```

11. Write a program to demonstrate Method Overriding where a Child class changes the behavior of a Parent class method.

File Name: OverrideDemo.java

Source Code:

```
class Parent {  
    void show() {  
        System.out.println("This is the parent class method.");  
    }  
}  
  
class Child extends Parent {  
    @Override  
    void show() {  
        System.out.println("This is the child class method.");  
    }  
}  
  
public class OverrideDemo {  
    public static void main(String[] args) {  
        Parent obj = new Child();  
        obj.show();  
    }  
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac OverrideDemo.java  
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java OverrideDemo  
This is the child class method.  
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```

12. Create a program that handles ArithmeticException and ArrayIndexOutOfBoundsException using multiple try-catch blocks and a finally block.

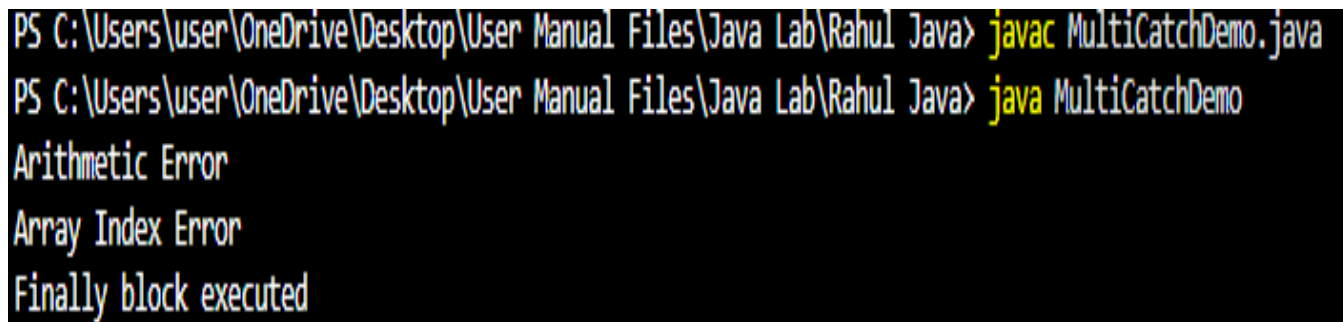
File Name: MultiCatchDemo.java

Source Code:

```
public class MultiCatchDemo {
    public static void main(String[] args) {
        try {
            int a = 10 / 0;
        } catch (ArithmeticException e) {
            System.out.println("Arithmetic Error");
        }

        try {
            int[] arr = new int[2];
            arr[5] = 10;
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Array Index Error");
        } finally {
            System.out.println("Finally block executed");
        }
    }
}
```

Output:



```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac MultiCatchDemo.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java MultiCatchDemo
Arithmetic Error
Array Index Error
Finally block executed
```

13. Write a program to create a Custom Exception class called InvalidAgeException and trigger it using the throw keyword.

File Name: CustomExceptionDemo.java

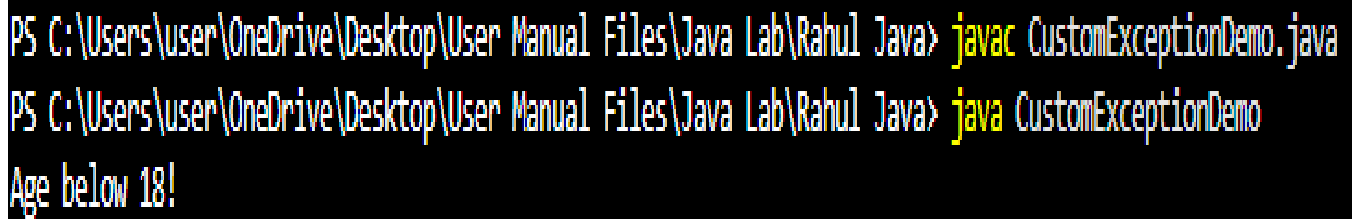
Source Code:

```
class InvalidAgeException extends Exception {
    InvalidAgeException(String message) {
        super(message);
    }
}

public class CustomExceptionDemo {
    static void check(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException("Age below 18!");
        }
        System.out.println("Valid Age");
    }

    public static void main(String[] args) {
        try {
            check(16);
        } catch (InvalidAgeException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Output:



```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac CustomExceptionDemo.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java CustomExceptionDemo
Age below 18!
```

14. Create a Multithreaded Program by extending the Thread class that prints numbers with a 1-second delay using Thread.sleep().

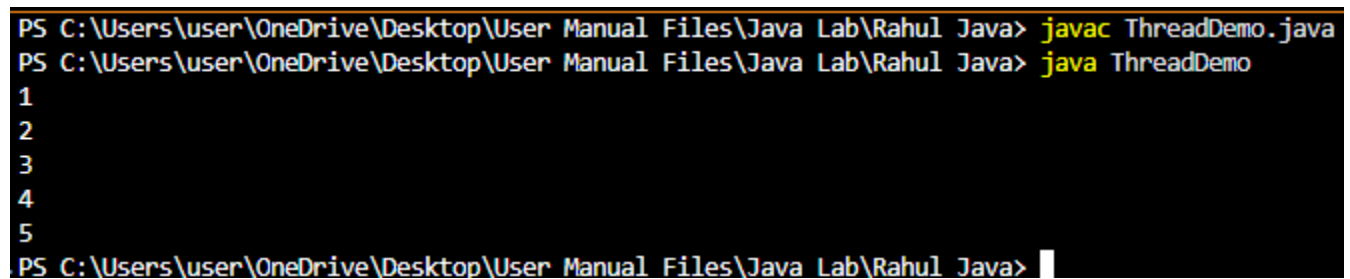
File Name: ThreadDemo.java

Source Code:

```
class MyThread extends Thread {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            System.out.println(i);
            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {
                System.out.println("Thread interrupted");
            }
        }
    }
}

public class ThreadDemo {
    public static void main(String[] args) {
        new MyThread().start();
    }
}
```

Output:



```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac ThreadDemo.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java ThreadDemo
1
2
3
4
5
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java>
```

15. Write a program to perform File Copy operations using FileInputStream and FileOutputStream to handle image or text data.

File Name: FileCopy.java

Source Code:

```
import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;

public class FileCopy {
    public static void main(String[] args) throws IOException {
        File src = new File("source.txt");
        FileOutputStream fos = new FileOutputStream(src);
        fos.write("Hi, I am Rahul!".getBytes());
        fos.close();

        FileInputStream fis = new FileInputStream("source.txt");
        FileOutputStream out = new FileOutputStream("destination.txt");
        int b;
        while ((b = fis.read()) != -1) {
            out.write(b);
        }
        fis.close();
        out.close();
        System.out.println("File copied successfully.");
    }
}
```

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac FileCopy.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java FileCopy
File copied successfully.
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> 
```



Rahul Java > destination.txt

1 Hello, I am Rahul!

Output:

16. Write a program to read data from a text file using FileReader and BufferedReader, and count the total number of lines in the file.

File Name: CountLines.java

Source Code:

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;

public class CountLines {
    public static void main(String[] args) {
        int count = 0;

        try {
            BufferedReader br = new BufferedReader(
                new FileReader("source.txt")
            );

            while (br.readLine() != null) {
                count++;
            }

            br.close();

            System.out.println("Total Lines = " + count);

        } catch (IOException e) {
            System.out.println("File error: " + e.getMessage());
        }
    }
}
```

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac CountLines.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java CountLines
Total Lines = 2
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```

Output:

```
Rahul Java > source.txt
1 Hello, I am Rahul!
2 This is line TWO.
3
```


17. Write a program to demonstrate Object Serialization by saving a Student object to a file and then retrieving it using Object Deserialization.

File Name: SerializeDemo.java

Source Code:

```
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.ObjectInputStream;
import java.io.ObjectOutputStream;
import java.io.Serializable;

class Student implements Serializable {
    private static final long serialVersionUID = 1L;

    String name;

    Student(String name) {
        this.name = name;
    }
}

public class SerializeDemo {
    public static void main(String[] args) {
        try {
            Student student = new Student(
                "Hello, I am Rahul from Kathmandu"
            );

            ObjectOutputStream output = new ObjectOutputStream(
                new FileOutputStream("student.dat"));
            output.writeObject(student);
            output.close();
            System.out.println("Student object saved successfully.");
            ObjectInputStream input = new ObjectInputStream(
                new FileInputStream("student.dat")
            );
            Student retrievedStudent = (Student) input.readObject();
            input.close();

            System.out.println("Retrieved student information:");
            System.out.println(retrievedStudent.name);

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Output:

```
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> javac SerializeDemo.java
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> java SerializeDemo
Student object saved successfully.
Retrieved student information:
Hello, I am Rahul from Kathmandu
PS C:\Users\user\OneDrive\Desktop\User Manual Files\Java Lab\Rahul Java> |
```